

AI-POWERED URBAN CANOPY HEALTH PLATFORM

TreeDoctor

"Google Maps tells us where trees are. TreeDoctor tells us which ones need our help."

An intelligent early warning and clinical EHR platform combining Multimodal Vision AI (Gemini 2.5 Flash), longitudinal time-series health tracking, real-time microclimate atmospheric telemetry, and automated municipal field dispatch.

VISION ENGINE

Gemini 2.5 Flash

EARLY WARNING

Longitudinal EHR Alerts

ATMOSPHERIC GRID

Open-Meteo & Copernicus

Urban Trees Are Dying in Plain Sight

Cities spend millions planting trees, yet lose up to 30% prematurely due to a total lack of clinical health visibility.

BLIND SPOT

Static & Stale Inventories

Municipal registries are static spreadsheets or GIS pins updated only once every 5 to 7 years.

Tree pathogens, drought stress, and root decay progress in weeks, rendering municipal records obsolete.

FINANCIAL DRAIN

The Reactive Cost Trap

Cities intervene only after a tree collapses on a sidewalk or power line.

Removal and replanting costs over \$3,200 per tree, vs. just \$45 for targeted preventative treatment caught 60 days earlier.

CLIMATE MULTIPLIER

Environmental Stress

Urban heat islands (>38°C), asphalt root compaction, vapor pressure deficits, and PM2.5 pollution starve canopies before outward symptoms become obvious to the naked eye.

From Static Inventory to Continuous Clinical Care

TreeDoctor transforms urban forestry into an automated Electronic Health Record (EHR) ecosystem.

01

Multimodal AI Vision

Gemini 2.5 Flash

Instant arborist pathology from field photos. Quantifies foliar necrosis, canopy thinning percentage, fungal conks, and pest boring holes with a clinical 0-100 score.

02

Longitudinal EHR Alerts

Time-Series Intelligence

Maintains historical health curves for each specimen. Automatically fires urgent alerts whenever a tree experiences an acute drop (>10 health points) between scans.

03

Microclimate Grounding

Zero-Mock Telemetry

Cross-checks visual decline against real-time ambient heat, 7-day rainfall deficits, relative humidity, and Copernicus AQI (PM2.5/PM10) at exact GPS coordinates.

04

Municipal Field Dispatch

Closed-Loop Action

Converts diagnoses into municipal work orders (deep-root irrigation, micro-injection, pruning) with full lifecycle tracking from Pending to Resolved.

Stakeholders, Pain Points & Value Delivered

Empowering public works departments, certified arborists, university campuses, and civic volunteers.

CITY GOV

Municipal Parks & Forestry Departments

Pain Point: Stretched field teams, emergency tree failures, slow manual audits, citizen complaint backlogs.

Value: Real-time city canopy health map, early warning decline alerts, and prioritized automated field dispatch.

COMMERCIAL

Certified Arborists & Tree Contractors

Pain Point: Time-consuming manual paper assessments, difficult client reporting, lack of objective longitudinal records.

Value: AI-assisted pathology scoring (0-100), digital audit trails, and photographic time-series comparison.

CAMPUSES

University Campuses & Botanical Gardens

Pain Point: High-value heritage specimens vulnerable to microclimate heat islands and localized pathogen outbreaks.

Value: Micro-location weather telemetry cross-referencing rainfall deficits to automate preventative care.

COMMUNITY

Civic Volunteers & Citizen Stewards

Pain Point: Want to protect local neighborhood trees but lack botanical expertise and direct reporting lines to city hall.

Value: Zero-friction camera scanner, instant AI diagnosis, and one-tap community reporting into the city queue.

Production-Grade Full-Stack & Multimodal Architecture

Secure server-side AI execution, zero-mock atmospheric telemetry, and high-performance geospatial UI.

AI & VISION LAYER

Gemini 2.5 Flash

- Multimodal vision analysis of leaves, bark, and canopy.
- Pathology identification: chlorosis, anthracnose, bracket fungi, borers.
- Structured clinical JSON: 0-100 score, urgency & treatment steps.
- Strict server-side isolation: API keys never exposed to client browser.

ATMOSPHERIC TELEMETRY

Open-Meteo & Copernicus

- Zero-mock live telemetry grounded to exact tree GPS coordinates.
- Ambient temperature and 7-day cumulative rainfall deficit analysis.
- Relative humidity & Vapor Pressure Deficit (VPD) evaluation.
- Copernicus Air Quality Index (AQI) with PM2.5 & PM10 particulate levels.

FRONTEND & SERVER

React 19 + Express + Leaflet

- Zero-typing tree registration with interactive Leaflet map pinning.
- 3 base map modes: Dark Canvas, Satellite Imagery, Topographic Terrain.
- Longitudinal SVG charts with decline detection threshold (>10 pts).
- Node.js Express backend bundled via esbuild into single CommonJS file.

The \$5.8B Opportunity & Quantifiable Municipal ROI

Saving public funds, reducing infrastructure hazard liability, and protecting mature carbon assets.

QUANTIFIABLE MUNICIPAL IMPACT

\$250,000+

Annual savings per 100 rescued trees vs. removal and replanting costs.

60% Drop

Reduction in young sapling mortality through proactive microclimate hydration.

70x Carbon

Mature trees sequester up to 70x more carbon than newly planted saplings.

\$5.8 Billion

Global Smart City Urban Greening & Tree Inventory Market by 2030.

B2G Municipal SaaS

PRIMARY REVENUE

Annual tiered subscription for city parks and public works departments (\$15,000 to \$80,000/year based on tree population volume, GIS polygon tiers, and crew dispatch seats).

Enterprise Campus Licensing

HIGH MARGIN

Dedicated canopy monitoring for corporate HQs, golf resorts, university campuses, and botanical arboretums protecting irreplaceable heritage assets.

Insurance & ESG Carbon Verification API

DATA MONETIZATION

Risk data feeds for municipal property insurers to evaluate branch collapse liabilities, and certified carbon telemetry for urban ESG green-bond credits.

Architected to Expand from Single Parks to Millions of Trees

High-throughput geospatial indexing, edge diagnostics, and seamless GIS software interoperability.

■ Geospatial Partitioning & Quadtree Clustering

Leaflet viewport-bounded queries and spatial index clustering ensure sub-second map rendering even with hundreds of thousands of specimens across metropolitan zones.

■ Decoupled Diagnostic Pipeline

Asynchronous image compression and queue workers prevent bottlenecks during city-wide citizen audit campaigns, maintaining rapid responses under high load.

■ Interoperable Forestry Standards

Data schemas align with USDA Forest Service & ISA arboricultural standards, enabling two-way sync with legacy municipal tools like Esri ArcGIS and TreePlotter.

■ Crowdsourced Force Multiplier

By turning every smartphone into an arborist scanner, city inspection bandwidth multiplies by 100x without increasing public municipal headcount.

What We Would Build Next with More Time & Resources

Satellite NDVI remote sensing, on-device Gemini Nano, IoT ground sensors, and predictive epidemiology.

PHASE 1

Satellite NDVI

Sentinel-2 & Planet Labs

Auto-detect canopy moisture stress and chlorophyll reduction from orbit using Normalized Difference Vegetation Index (NDVI) before field teams even deploy.

PHASE 2

Mobile PWA + Nano

Zero-Connectivity Mode

Deploy on-device Gemini Nano / quantized edge vision models directly onto park ranger devices for zero-reception national parks and nature reserves.

PHASE 3

IoT Ground Probes

LoRaWAN Smart Irrigation

Sub-surface soil sensor probes that trigger automated municipal drip valves when soil water tension drops below species-specific wilting points.

PHASE 4

Predictive Models

Epidemiological Vectors

Machine learning forecasting of pathogen spread trajectories (e.g. Emerald Ash Borer) by cross-referencing wind corridors and tree species density.